

Practical Homework 2: Support Vector Machines

50 Points Possible

4/26/2026

Attempt 1



In Progress

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Unlimited Attempts Allowed**Details**

Overview and Goals

The second practical homework will explore the use of support vector models for classification. Using the concepts from our two lectures on support vector models, your task is to explore how health behaviors impact health outcomes. The data was collected via the National Health Interview Survey and was accessed through IPUMS Health Survey¹. This data set contains variables regarding respondent's demographics, 5 major health conditions such as (cancer, heart disease, diabetes, heart attack and stroke) and their lifestyle and behaviors like work hours, physical activity, sleep habits, and eating habits. Details about the specific variables and the key to decoding the meaning of numeric values for each variable can be found in the codebook.

Your task is to predict the presence of one of these 5 diseases based on demographics and habits.

You should explore and be creative with your answers. The main task is to perform the prediction, but you should also interrogate the data and pose questions that are interesting to you.

Deliverables

This homework assignment has two deliverables: your code and a poster describing your work and results.

Code Requirements (50%)

You are welcome to subset the data in any way you like, e.g. only look at married couples, or only focus on children, but you should investigate the relationships among at least **5 total variables and the 3 models types (linear, radial, and polynomial kernels)**. The 5 variables do not each have to be in every model.

As before, your code must be clean and understandable, including comments and organization that will help the reader/grader understand your thought process. Your work should be saved in a GitHub repository and your submission should be the public URL where all your work can be found.

Communication Requirements (50%)

For this assignment, your deliverable is a poster. Your poster should include the information required in the [standard content](https://seattleu.instructure.com/courses/1632903/files/73438171?wrap=1) (<https://seattleu.instructure.com/courses/1632903/files/73438171?wrap=1>). 

(https://seattleu.instructure.com/courses/1632903/files/73438171/download?download_frd=1) and should be attractively designed and informative. Space on the poster is limited, so this assignment will assess your ability to discern the important information and communicate it in a concise way. Consider the audience to be a colleague or client with mathematical expertise, but who is unfamiliar with this particular modeling framework. Perhaps a better way to think about it: make this poster for *yourself* a few years into the future. Assume you can recall the fundamentals, but the details of the method are lost.

The easiest way to make your poster is using a presentation software like PowerPoint, Keynote or Google Slides. It is recommended that you size your document to be 36" x 48" and use a **minimum** font size of 24 pt for the body of the poster. It is strongly encouraged to use

graphics, plots, and images. Any diagrams or plots must be custom made (e.g. don't pull an example of a SVM from a blog), but can use icons/images as decoration if cited.

Technical Background: In your technical background, you must fully describe a support vector machine, including the relevant equations, tuning parameters and considerations for its application. However, you are not expected to discuss the precursor/simpler models to save on space.

Plot: At least one plot of an example SVM decision boundary on two strong predictor variables. This can be a "cartoon" plot or a plot of made up data, but you should demonstrate the type of plot and classifications you are creating.

Discussion/Conclusions: Discuss which variables seemed to be strong predictors of disease. Given the interpretation of the SVM plots, what does this say about the habits or metrics that may impact diseases? If any demographic or social factors are used as predictors, how does this make sense? If you were to make a suggestion to policy-makers, what would you say, based off your findings?

You must include the citation from the bottom of this assignment in your poster.

Dataset

This data set and codebook can be [downloaded from my GitHub](https://github.com/mendible/5322/tree/main/Homework%202)  [. \(https://github.com/mendible/5322/tree/main/Homework%202\)](https://github.com/mendible/5322/tree/main/Homework%202). It contains mostly quantitative predictors, such as number of hours or frequency of a behavior. I provide here 48 variables generally one of 4 themes:

- **Survey information (variables 1-12):** These variables are ID numbers or sample weights from the survey methodology. You can largely ignore these for this homework, but know that this is not a representative sample for the United States-- weighting is important to make extrapolating claims.
- **Demographic information (variables 13-18, 22):** Some of these variables can be used as predictors, but not all. Again, be sure to carefully recode these to make sense for how you apply them. You may want to compare how your models work for people in different groups (e.g. split by sex or marital status).
- **Disease indicators (variables 23-27):** Indicator of simply whether or not someone has ever had any of the 5 diseases of interest. One or more of these should be your target variable.
- **Basic health metrics and habits (all others):** All measurements of health and habits, such as amounts of physical activity, frequency of eating certain indicator foods like vegetables or pizza, etc. Note that while most of these codes are identical to the number they represent, they still need to be carefully cleaned!

Size: The data set large and you are welcome to subset it to reduce the size (e.g. only consider adults or only use responses from the Northeast region).

Categorical Data: Recall that we need to use hyperplanes to divide all our data. Be very careful about numerically-coded categorical data here.

[1] Lynn A. Blewett, Julia A. Rivera Drew, Miriam L. King, Kari C.W. Williams, Daniel Backman, Annie Chen, and Stephanie Richards. IPUMS Health Surveys: National Health Interview Survey, Version 7.4 [dataset]. Minneapolis, MN: IPUMS, 2024. <https://doi.org/10.18128/D070.V7.4>.  [. \(https://doi.org/10.18128/D070.V7.4\)](https://doi.org/10.18128/D070.V7.4) <http://www.nhis.ipums.org>  [. \(http://www.nhis.ipums.org\)](http://www.nhis.ipums.org).

View Rubric

Practical Homework				
Criteria	Ratings			Points
Title and Introduction view longer description	Full Marks 2 pts	Needs improvement 1 pts	No Marks 0 pts	/2 pts
Theoretical Background view longer description	Full Marks 4 pts	Needs improvement 2 pts	No Marks 0 pts	/4 pts

Criteria	Ratings			Points
Methodology view longer description	Full Marks 3 pts	Needs improvement 1.5 pts	No Marks 0 pts	/3 pts
Results view longer description	Full Marks 3 pts	Needs improvement 1.5 pts	No Marks 0 pts	/3 pts
Discussion view longer description	Full Marks 3 pts	Needs improvement 1.5 pts	No Marks 0 pts	/3 pts
Conclusions view longer description	Full Marks 1 pts	No Marks 0 pts		/1 pts
Overall Communication Style view longer description	Full Marks 2 pts	Needs improvement 1 pts	No Marks 0 pts	/2 pts
Specific Criteria view longer description	Full Marks 2 pts	Needs improvement 1 pts	No Marks 0 pts	/2 pts
Effort and Completion view longer description	Full Marks 5 pts	Needs Improvement 2.5 pts	No Marks 0 pts	/5 pts
Code: Data Prep view longer description	Full Marks 5 pts	Needs Improvement 2.5 pts	No Marks 0 pts	/5 pts
Code: Tuning view longer description	Full Marks 5 pts	Needs Improvement 2.5 pts	No Marks 0 pts	/5 pts
Code: Model Evaluation view longer description	Full Marks 5 pts	Needs Improvement 2.5 pts	No Marks 0 pts	/5 pts
Code: Model Comparison and Interpretation view longer description	Full Marks 5 pts	Needs Improvement 2.5 pts	No Marks 0 pts	/5 pts
Code: Organization and Clarity view longer description	Full Marks 5 pts	Needs Improvement 2.5 pts	No Marks 0 pts	/5 pts

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